SMART PLACEMENT TRACKER SYSTEM

USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

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in partial fulfilment of the End Semester Examination for the course

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "SMART PLACEMENT TRACKER SYSTEM" is the bonafide work of **SOWMYA R (241501207)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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Submitted for the End Semester Examination (Individual Project) held in October 2026.

INTERNAL EXAMINER	**EXTERNAL EXAMINER**

DECLARATION

I, **SOWMYA R (241501207)**, hereby declare that the project report titled "SMART PLACEMENT TRACKER SYSTEM", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data components, forms, reports and dashboard — was developed using the low-code application development features of Zoho Creator and has been reviewed and documented by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: October 2026

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SOWMYA R

ABSTRACT

Campus placement preparation requires students to coordinate profiles, job openings, eligibility criteria, applications, assessments, interviews, preparation tasks, certifications and skill requirements. When these are kept in separate spreadsheets, e-mails and notes, deadlines are missed and progress cannot be reviewed as a whole. Smart Placement Tracker is a low-code application built on Zoho Creator that brings these activities into one system for students and placement administrators.

The application is organised as eight data components — Student Profiles, Company Directory, Job Opportunities, Job Applications, Interview Scheduler, Preparation Tracker, Certification Tracker and Skill Analysis — presented through forms, list reports, a navigation menu and a dashboard. Job Applications reference students and job opportunities through lookup fields and record the application status, assessment date, interview date and notes. Three Deluge form-validation scripts enforce required fields and date-consistency rules at submission. The dashboard presents summary cards (total salary packages, average minimum CGPA, average completion percentage and average CGPA), job-status counts for Open, On Hold, Hired and Closed positions, quick links for data entry, and charts of planned against actual study hours and of task status.

The report documents the requirements, design, implementation, test cases and governance of the application. All fourteen test cases defined for the application passed. The application holds demonstration data, so dashboard values illustrate its behaviour and are not institutional placement statistics. Role-based permissions restrict each role to its authorised records; reminders, an activity log and external job-portal integration are identified as future work.

Keywords: Low-Code Development; Zoho Creator; Student Placement; Job Application Tracking; Skill Gap Analysis; Dashboard.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Smart Placement Tracker System using Zoho Creator
Application Domain	Education – Campus Placement Management
Student Name and Register No.	SOWMYA R – 241501207
Zoho Creator Edition	Student Edition
Application Name	Student Placement Management System (project title: Smart Placement Tracker)
Published / Live URL	Private; accessed through the developer's Zoho Creator account
Demo Credentials (per role)	Not reproduced in this report for security
Number of Forms	8 – Student Profile, Company Directory, Job Opportunity, Job Application, Interview Scheduler, Preparation Tracker, Certification Tracker, Skill Analysis
Number of Reports	List reports for each of the eight components
Number of Pages / Dashboards	1 Dashboard
Number of Workflows	3 Deluge validation scripts (Appendix C)
Integrations	None
Roles and Profiles Defined	Administrator, Student
Environment	Zoho Creator cloud-hosted application

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CGPA	Cumulative Grade Point Average
CO	Course Outcome
Deluge	Data Enriched Language for the Universal Grid Environment
ER	Entity–Relationship
FK	Foreign Key
FR	Functional Requirement
NFR	Non-Functional Requirement
PK	Primary Key
RBAC	Role-Based Access Control
TC	Test Case
UC	Use Case
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Background

Students preparing for campus placements must keep track of job openings, eligibility criteria, application dates, assessments, interviews, preparation tasks, certifications and technical skills. When this information is held in separate spreadsheets, e-mails and personal notes, it becomes difficult to identify deadlines, monitor the progress of each application and decide which skills need further work.

Smart Placement Tracker addresses this problem through a centralised low-code application built on Zoho Creator. It combines student profiles, company and job-opportunity records, application tracking, interview scheduling, preparation planning, certification records and skill analysis in a single browser-based interface.

1.2 Problem Statement

Placement-related information is usually spread across different files and platforms. Students may miss application deadlines, lose track of the stage reached in each application, or lack a clear view of the skills required for a chosen job role. Placement staff face the same fragmentation when they try to review applications and report on placement activity. The specific difficulties addressed by this project are:

- Application details are scattered across spreadsheets, e-mails and messages.
- Eligibility requirements and job information are difficult to compare consistently.
- Assessment and interview dates are easily overlooked without a consolidated tracker.
- Preparation tasks and certification details are maintained separately from applications.
- Students do not have a single summary of required skills, skills already possessed and skills still missing.

1.3 Objectives

The objectives of the project are identified as OBJ-1 to OBJ-5 and are evaluated in Chapter 8.

- **OBJ-1:** To centralise student profiles, company details and job-opportunity information.
- **OBJ-2:** To record job applications and track their status, assessment dates, interview dates and notes.
- **OBJ-3:** To organise interview scheduling, preparation priorities and certification records.
- **OBJ-4:** To present placement summaries through dashboard cards, status indicators and charts.
- **OBJ-5:** To help identify skill gaps by comparing the skills required for a job role with the skills a student already possesses.

1.4 Scope of the Project

The project covers the design and configuration of a Zoho Creator application with eight data components (Student Profiles, Company Directory, Job Opportunities, Job Applications, Interview Scheduler, Preparation Tracker, Certification Tracker and Skill Analysis), list reports for each component, a navigation menu, a dashboard and Deluge form-validation scripts for application and interview data.

The following items are outside the scope of this version and are listed as future enhancements in Section 9.2: automated reminders and notifications, integration with external job portals, resume parsing, an activity log, and a mobile-specific interface. The application is populated with demonstration data and is not connected to institutional placement records.

1.5 Justification for a Low-Code Approach

A low-code platform allows forms, reports, navigation and dashboards to be created visually, with custom code needed only where the standard components are not sufficient [1]. Zoho Creator suits this project because placement information is naturally organised as structured records that relate to one another, and because the platform supplies hosting, user interface components and a scripting language,

Deluge, for custom validation [3], [4]. Practitioner studies report that low-code development is easy to learn and fast for workflow-oriented applications, while also noting platform limits and customisation constraints [2]; these considerations are discussed in Section 8.3.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirements and use cases, Chapter 4 the system design and Chapter 5 the implementation. Chapter 6 describes the testing approach, test cases and results. Chapter 7 covers deployment, security and governance. Chapter 8 discusses the results against the objectives, and Chapter 9 gives the conclusion and future enhancements. The appendices contain the feature-coverage summary, course-outcome mapping, Deluge source code, user manual and the evaluation and viva-voce record sheets.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the conventional process, students use spreadsheets, e-mails, calendars and separate websites to track job opportunities, applications, interviews, preparation activities and certifications. Information must be updated manually in several places, which makes it hard to maintain a complete view of placement progress.

2.2 Limitations of the Existing System

- Information is distributed across files and platforms, which increases the chance of outdated or duplicated records.
- Application status, assessment dates and interview details are not visible in one place.
- Preparation progress, certifications and skill gaps are tracked separately from applications.
- There is no consolidated summary that the student or the placement office can review at a glance.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the effort involved in building the proposed application with a traditional approach and with Zoho Creator.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code)	Low-Code on Zoho Creator
Development effort	Separate design of the database, back end and user interface.	Forms, reports and dashboard built directly with the Zoho Creator application tools; Zia, the built-in AI assistant, drafts the initial layout.
Skills required	Programming, database design and web development.	Data modelling and Zoho Creator concepts; Deluge only where custom logic is needed [3].
Hosting	Server and deployment set-up and maintenance.	Hosted by the Zoho Creator platform.

Criterion	Traditional (Pro-Code)	Low-Code on Zoho Creator
Maintenance	Code changes, testing and redeployment.	Forms and views modified directly in the builder.
Customisation limits	Few limits on design choices.	Bound by platform components; custom logic needs Deluge [4].

2.4 Literature Survey

Table 2.2 summarises the sources that informed the project. The numbers in brackets refer to the References section.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source, Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021 [1]	"Low-Code Platform"	Relates low-code platforms to citizen developers, agility and productivity.	General background for choosing low-code (Section 1.5).
2	Y. Luo et al., 2021 [2]	"Characteristics and Challenges of Low-Code Development"	Analysis of 73 Stack Overflow and 228 Reddit posts: low-code is easy to learn and fast, suited to automated workflows; views on limitations are mixed.	Supports the justification (Section 1.5) and the limitations (Section 8.3).
3	Zoho Corporation [3], [4]	Deluge scripting (Zoho Creator and Deluge Help)	Deluge adds custom logic to Zoho applications.	Mechanism for the validation scripts in Appendix C.
4	U. Frank, P. Maier and A. Bock, 2021 [5]	Low Code Platforms: Promises, Concepts and Prospects	Compares ten low-code platforms using a common framework.	Places Zoho Creator among comparable platforms.

2.5 Proposed System

The proposed system is Smart Placement Tracker. Student profiles and the company directory provide supporting information; job opportunities store roles and eligibility criteria; job applications record each student's progress through the selection process; the interview scheduler and preparation tracker organise follow-up activities; the certification tracker maintains credential details; and skill analysis presents required

skills, skills already possessed and missing skills. The dashboard summarises selected placement indicators from the records entered.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Two user roles are defined for the application, as summarised in Table 3.1.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Student	A student who uses the tracker to organise placement preparation.	Review opportunities, maintain permitted profile information, record and track applications and interviews, manage preparation tasks and certifications, and review skill gaps.
Placement Administrator	An authorised staff member who manages placement information.	Maintain company and job-opportunity information, review applications and interview schedules, update placement status, review reports and manage access.

3.2 Functional Requirements

The functional requirements FR-01 to FR-10 are listed in Table 3.2. Section 5.4 maps each requirement to its implementation status and Chapter 6 maps each to one or more test cases.

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	Maintain student profile details: department, graduation year, CGPA and technical skills.	Student; Placement Administrator	High
FR-02	Create and review job-opportunity records with required skills and eligibility criteria.	Placement Administrator; Student (view)	High
FR-03	Record job applications with status, assessment date, interview date and notes, and allow the status to be updated.	Student; Placement Administrator	High

FR ID	Requirement	Role(s)	Priority
FR-04	Organise preparation tasks with priorities and planned and actual study hours.	Student	High
FR-05	Maintain certification records.	Student	Medium
FR-06	Compare job-role requirements with possessed and missing skills.	Student; Placement Administrator	High
FR-07	Display placement summaries, status counts and charts on a dashboard.	Student; Placement Administrator	Medium
FR-08	Schedule interviews and validate interview and assessment dates.	Student; Placement Administrator	High
FR-09	Maintain the company directory.	Placement Administrator	Medium
FR-10	Restrict access to data and functions according to the user role.	Placement Administrator	High

An activity log was considered in the early requirements; it is not part of the current application and is recorded as a future enhancement (Section 9.2).

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard should remain responsive for the expected volume of placement records.
NFR-02	Security	Personal and academic information should be visible only to authorised users.
NFR-03	Usability	Users should be able to navigate the modules and understand the application-status fields without training.
NFR-04	Availability	Availability depends on the Zoho Creator account and subscription plan.
NFR-05	Data quality	Records and dashboard calculations should be reviewed and kept up to date.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Student Edition)
Scripting language	Deluge, used for the form-validation scripts in Appendix C
Browser	Modern web browser such as Google Chrome or Microsoft Edge
Other services / API tools	None; no external integration is used
Client hardware	Computer with a modern browser and an internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases for the two actors. Students maintain their profile, view and apply for opportunities, track applications, manage preparation tasks, skills and certifications, and review skill gaps and the dashboard. Placement Administrators manage opportunities and company details, review applications, manage interview schedules, update placement status and review reports. Tracking an application includes recording assessment details, scheduling an interview and updating the selection status. Table 3.5 lists every use case shown in the figure and its related requirement.

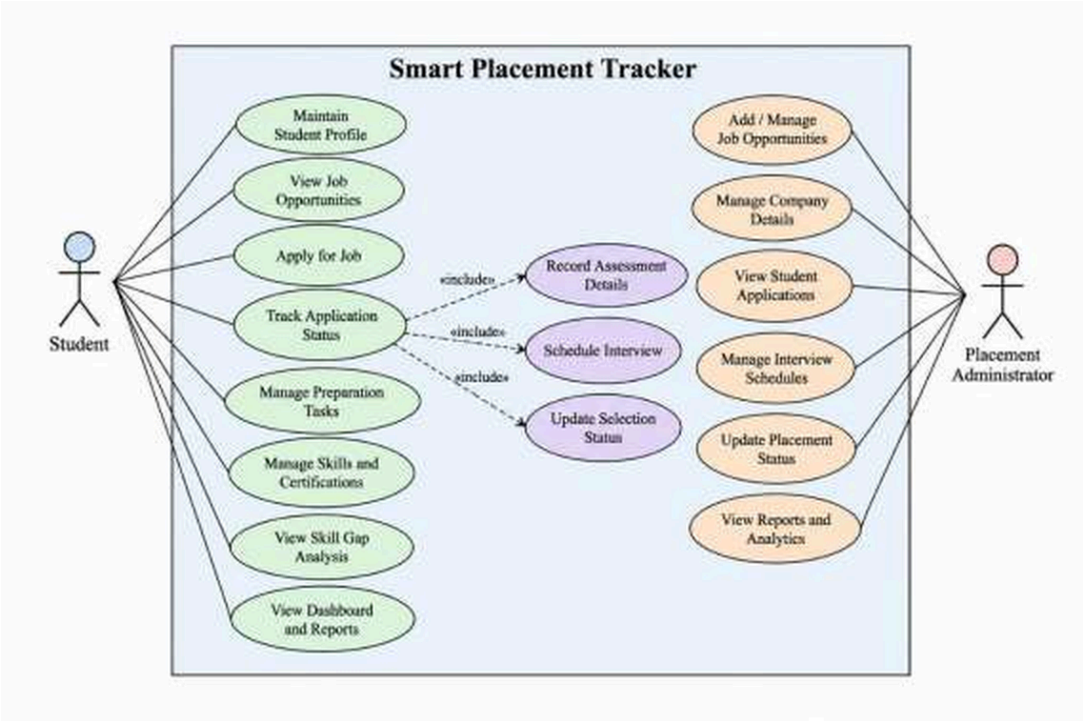


Figure 3.1: Use Case Diagram of the Smart Placement Tracker

Table 3.5: Summary of Main Use Cases

UC ID	Use Case	Actor	Related FR
UC-01	Maintain Student Profile	Student	FR-01
UC-02	View Job Opportunities	Student	FR-02
UC-03	Apply for Job	Student	FR-03
UC-04	Track Application Status	Student	FR-03
UC-05	Manage Preparation Tasks	Student	FR-04
UC-06	Manage Skills and Certifications	Student	FR-05
UC-07	View Skill Gap Analysis	Student	FR-06

UC ID	Use Case	Actor	Related FR
UC-08	View Dashboard and Reports	Student, Placement Administrator	FR-07
UC-09	Add / Manage Job Opportunities	Placement Administrator	FR-02
UC-10	Manage Company Details	Placement Administrator	FR-09
UC-11	View Student Applications	Placement Administrator	FR-03
UC-12	Manage Interview Schedules	Placement Administrator	FR-08
UC-13	Update Placement Status	Placement Administrator	FR-03
UC-14	View Reports and Analytics	Placement Administrator	FR-07
UC-15	Record Assessment Details (included in UC-04)	Student, Placement Administrator	FR-03
UC-16	Schedule Interview (included in UC-04)	Student, Placement Administrator	FR-08
UC-17	Update Selection Status (included in UC-04)	Placement Administrator	FR-03

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The system is accessed through a web browser. Zoho Creator supplies the application layer — forms, reports, the navigation menu, the dashboard and the Deluge validation workflows — while its hosted data layer stores the records of the eight data components. No external services are connected.

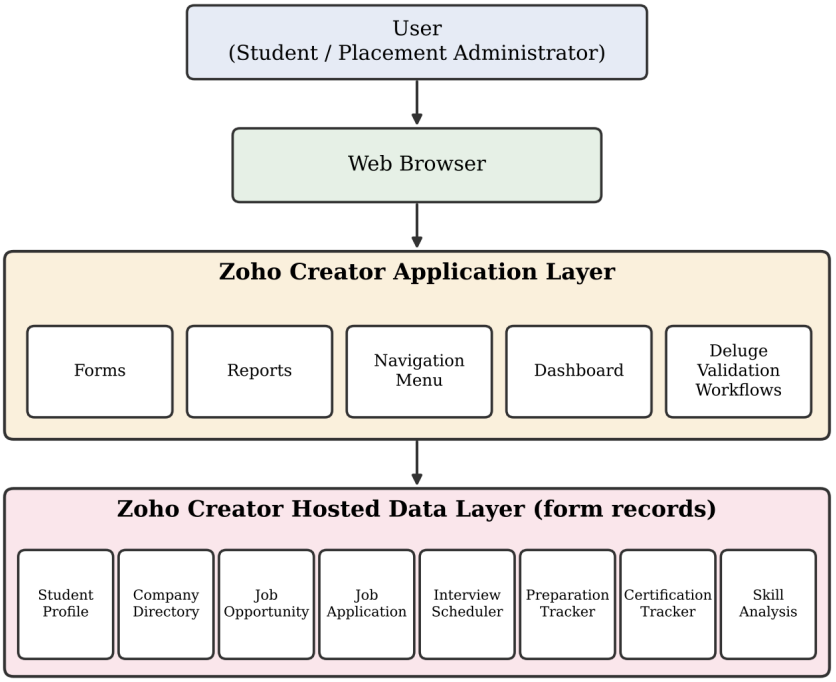


Figure 4.1: System Architecture of the Smart Placement Tracker

4.2 Data Model

Figure 4.2 shows the conceptual data model. A student submits many job applications, and each application refers to one job opportunity; a company offers many job opportunities; an application can lead to interview records; and a student maintains many preparation and certification items. Skill Analysis is a derived view that compares the required skills of a job opportunity with the skills held in the student profile.

In the Zoho Creator implementation, the Job Application form references Student Profile and Job Opportunity through lookup fields; this is visible on the form, where

Student Name and Job Opportunity are selected from existing records (Figure 5.2). The remaining associations in the figure describe how the records relate by student, company, job role or application. Company Name and Job Role on the Job Application form are text fields. Table 4.1 lists the data components and their relationships.

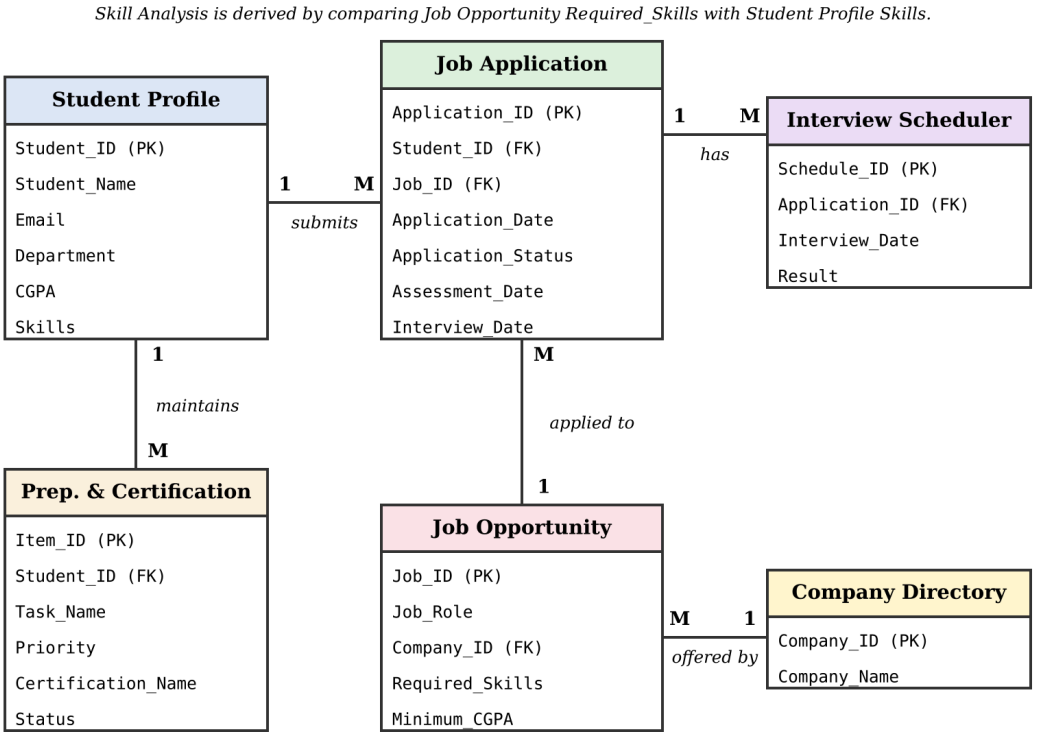


Figure 4.2: Conceptual Data Model of the Smart Placement Tracker

Table 4.1: Data Components and Their Relationships

Component	Purpose	Key Fields	Relationship
Student Profile	Maintain student academic and skill information.	Student Name, Email Address, Department, Graduation Year, CGPA, Technical Skills	Referenced by Job Application (lookup); linked to Preparation Tracker and Certification Tracker by student.
Company Directory	Maintain the companies that offer opportunities.	Company Name and company details	Offers Job Opportunities (by company name).
Job Opportunity	Store available roles and eligibility information.	Company Name, Job Title, Job Description, Required Skills, Minimum CGPA, Eligible Department	Referenced by Job Application (lookup).

Component	Purpose	Key Fields	Relationship
Job Application	Track a student's application for a role.	Student Name, Job Opportunity, Company Name, Job Role, Application Date, Application Status, Assessment Date, Interview Date, Notes	Lookups to Student Profile and Job Opportunity.
Interview Scheduler	Organise interview details.	Student Name, Company Name, Job Role, Interview Date	Linked to Job Application by student, company and role.
Preparation Tracker	Organise preparation tasks and priorities.	Task or subject, Student Name, Priority (High / Medium / Low), Task Status, Planned and Actual Study Hours, Completion Percentage	Linked to Student Profile by student.
Certification Tracker	Maintain certification details.	Certification details recorded per student	Linked to Student Profile by student.
Skill Analysis	Compare role requirements with skills possessed.	Student Name, Selected Job Role, Required Skills, Skills Already Possessed, Missing Skills, Matched Skills Count	Draws on Student Profile and Job Opportunity.

4.3 Form and Navigation Design

The navigation menu gives access to the Dashboard and to the modules listed in Table 4.2. Forms support data entry and list reports display the stored records. The Dashboard also provides quick links for adding a job application, job opportunity, preparation tracker and certification tracker.

Table 4.2: Navigation Sections, Forms and Views

Navigation Section	Data Component	Forms and Views
Dashboard	Summary view	Summary cards, job-status counts, quick links and charts
Job Applications	Job Application	Job Application form; All Job Applications report; Application Statuses view
Job Opportunities	Job Opportunity	Job Opportunity form; All Job Opportunities report
Preparation Trackers	Preparation Tracker	Preparation Tracker form; priority view (High, Medium, Low)
Certification Trackers	Certification Tracker	Certification Tracker form and report

Navigation Section	Data Component	Forms and Views
Interview Scheduler	Interview Scheduler	Interview scheduling form and report
Student Profiles	Student Profile	Student Profile form; All Student Profiles report
Skill Analyses	Skill Analysis	All Skill Analyses report
Company Directory	Company Directory	Company form and report

4.4 Process and Workflow Design

The process, shown in Figure 4.3, begins with maintaining student and company information and adding job opportunities. When a student applies, a job application record is created and its status is updated as assessments and interviews take place. Preparation tasks and certifications are managed alongside the application, skill gaps are reviewed for the selected role, and the dashboard and reports summarise progress. Reminders and automatic status updates, shown as dashed links, are planned enhancements.

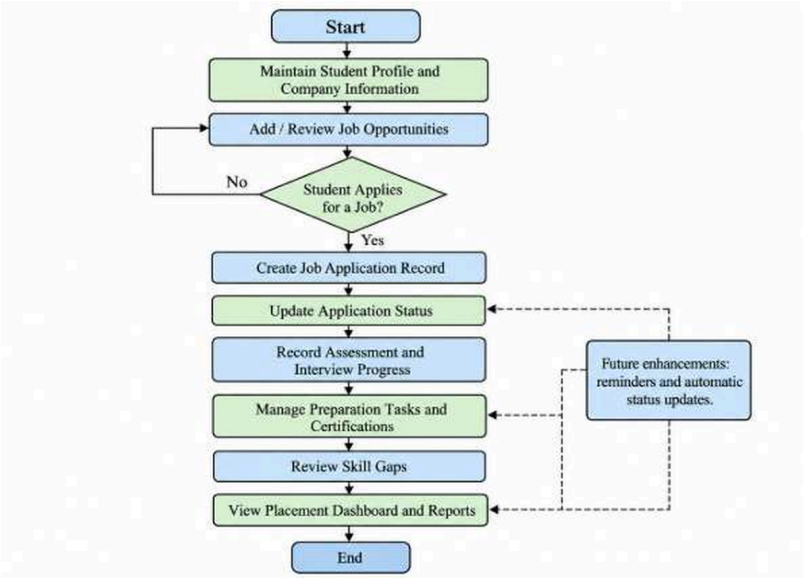


Figure 4.3: Placement Tracking Process

Data entry is protected by three Deluge validation scripts that run when a form is submitted. They stop the submission and show a message when required values are missing or when dates are inconsistent. Table 4.3 summarises their rules; the complete code is given in Appendix C.

Table 4.3: Deluge Validation Workflows

Script	Form	Trigger	Rules Enforced
C.1	Job Application	On submit (validate)	Assessment Date and Interview Date must not be earlier than Application Date.
C.2	Interview Scheduler	On submit (validate)	Student Name, Company Name, Job Role and Interview Date are required; Interview Date must not be in the past.
C.3	Job Application	On submit (validate)	Student Name and Application Date are required; Interview Date must not be before Application Date.

4.5 Access Control Design

Access is designed around the two roles in Table 3.1. Students manage their own profile and placement records and may view shared reference data such as job opportunities; Placement Administrators have broader access to maintain reference data and review all records. Table 4.4 records the intended permissions, where A = add, E = edit, V = view and D = delete. The permissions are configured through the Roles and Permissions settings of Zoho Creator and verified by test case TC-12 (Section 7.2).

Table 4.4: Role–Permission Matrix

Component	Placement Administrator	Student
Student Profile	A, E, V	V, E (own record)
Company Directory	A, E, V, D	V
Job Opportunity	A, E, V, D	V
Job Application	A, E, V	A, V, E (own records)
Preparation Tracker	A, E, V	A, V, E (own records)
Certification Tracker	A, E, V	A, V, E (own records)
Interview Scheduler	A, E, V	V (own records)

Component	Placement Administrator	Student
Skill Analysis and Dashboard	V	V

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in Zoho Creator as a single application. Its display name in the Zoho Creator workspace is “Student Placement Management System”; it is documented in this report under the project title Smart Placement Tracker. The forms, reports, navigation menu and dashboard were built and configured with the Zoho Creator application tools, with Zia, the built-in AI assistant, used to draft the initial layout of the forms and reports. The Deluge validation scripts of Appendix C were added to the forms.

The navigation menu provides the Dashboard and the modules Job Applications, Job Opportunities, Preparation Trackers, Certification Trackers, Interview Scheduler, Student Profiles, Skill Analyses and Company Directory. Figure 5.1 shows the menu alongside the dashboard.

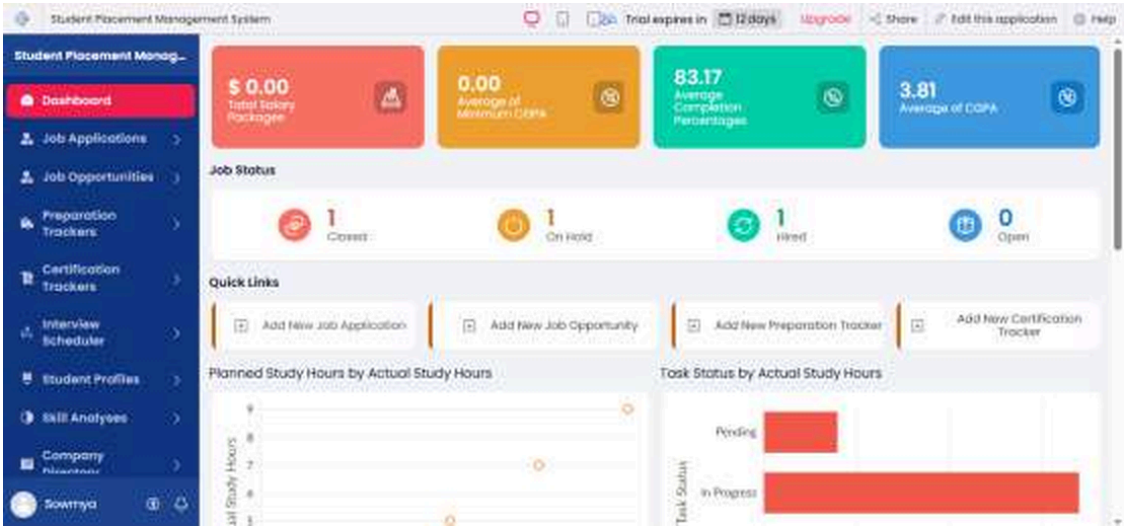


Figure 5.1: Smart Placement Tracker Navigation and Dashboard (Demonstration Data)

5.2 Forms and Data Capture

The Job Application form (Figure 5.2) is the central data-entry form. Student Name and Job Opportunity are lookup fields that select existing student and opportunity records; Company Name and Job Role are text fields; Application Date is a date field that defaults to the current date; Application Status is a drop-down; Assessment Date and Interview Date are date fields in the dd-MMM-yyyy format; and Notes is a

multi-line field. The form therefore captures everything needed to follow one application from submission to selection.

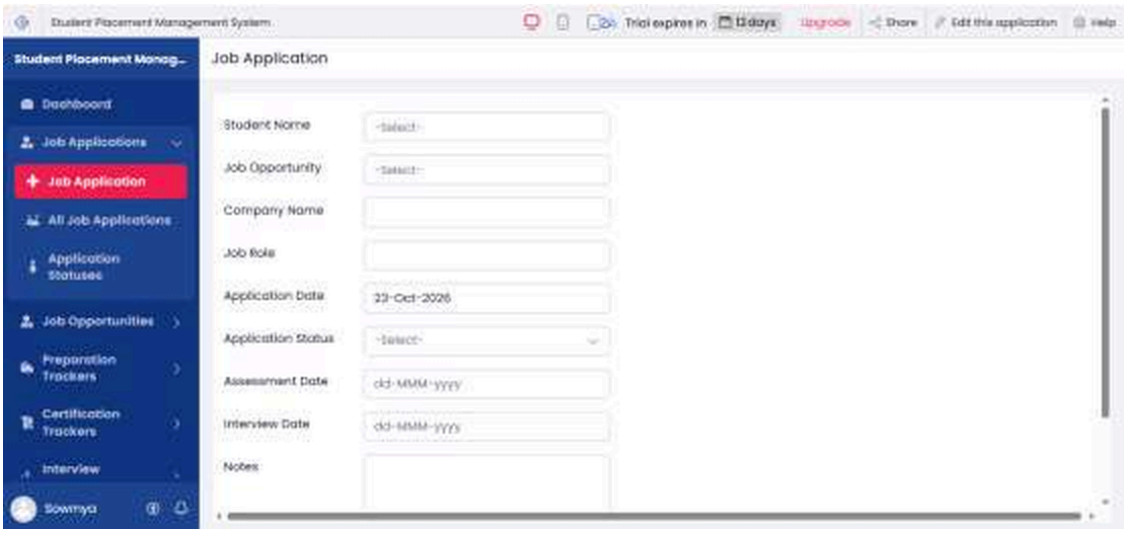


Figure 5.2: Job Application Form

The other forms follow the same pattern. The Student Profile form stores name, e-mail address, department, graduation year, CGPA and technical skills. The Job Opportunity form stores the company, job title, description, required skills, minimum CGPA and eligible department. The Preparation Tracker form stores the task, student, priority, task status and planned and actual study hours. The Certification Tracker and Interview Scheduler forms record certification details and interview schedules. Date rules on the Job Application and Interview Scheduler forms are enforced by the Deluge scripts described in Section 4.4.

5.3 Reports and Dashboard

Each module has a list report. The All Job Applications report (Figure 5.3) lists the application ID, student name, job opportunity, company name, job role, application date and status for every record. The Job Opportunities report lists company, role, required skills, minimum CGPA and eligible department. The Student Profiles report lists name, department, graduation year, CGPA and skills. The Skill Analyses report lists, for each student and selected job role, the required skills, the skills already possessed, the missing skills and the matched-skills count. The Preparation Tracker presents tasks in High, Medium and Low priority groups.

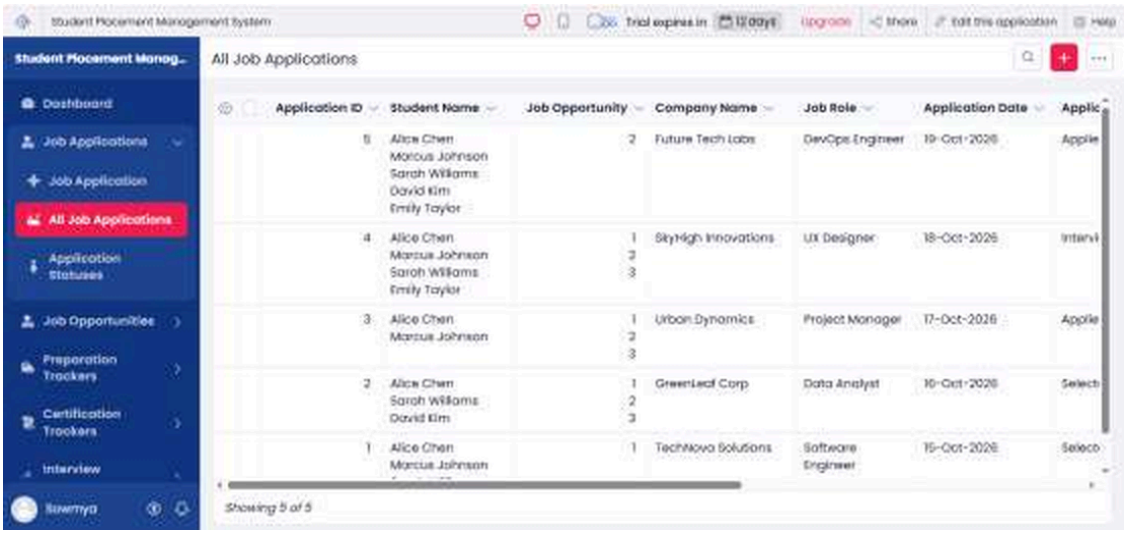


Figure 5.3: All Job Applications Report (Demonstration Data)

The dashboard (Figure 5.1) consolidates placement indicators, application-status counts, quick links and preparation analytics. It contains four summary cards, a Job Status panel with counts for Closed, On Hold, Hired and Open positions, quick links for adding a job application, job opportunity, preparation tracker and certification tracker, and two charts: planned study hours against actual study hours, and task status (Pending and In Progress) against actual study hours. Table 5.1 explains each measure.

The values on the dashboard are computed from demonstration records entered during development. They illustrate the behaviour of the application and must not be read as institutional placement statistics.

Table 5.1: Dashboard Measures

Measure	Meaning	Source Component	Value in Demonstration Data
Total Salary Packages	Aggregate (sum) of the salary package amounts recorded for job opportunities.	Job Opportunity	$ 0.00
Average of Minimum CGPA	Average of the minimum-CGPA eligibility values across job opportunities.	Job Opportunity	0.00
Average Completion Percentages	Average completion percentage across preparation records.	Preparation Tracker	83.17
Average of CGPA	Average CGPA across student profiles.	Student Profile	3.81
Job Status counts	Number of records in each status: Closed, On Hold, Hired, Open.	Job records	1, 1, 1, 1
Planned vs Actual Study Hours	Scatter chart comparing planned and actual hours for each preparation record.	Preparation Tracker	Chart
Task Status chart	Bar chart of Pending and In Progress tasks against actual study hours.	Preparation Tracker	Chart

The values of 0.00 for Total Salary Packages and Average of Minimum CGPA indicate that the demonstration job-opportunity records do not contain salary or minimum-CGPA values; the two cards show real values once those fields are populated (see I-02 in Section 6.4).

5.4 Implementation Details and Feature Status

Table 5.2 maps every requirement to the feature that implements it and records its status.

Table 5.2: Implementation Status Summary

Feature	Req.	Status	Remark
Student Profiles form and report	FR-01	Implemented	Name, e-mail, department, graduation year, CGPA and skills.
Job Opportunities form and report	FR-02	Implemented	Company, role, required skills, minimum CGPA and eligible department.
Job Application form and report	FR-03	Implemented	Lookups to student and opportunity; status, assessment and interview dates, notes.
Preparation Tracker and priorities	FR-04	Implemented	High, Medium and Low priority groups; study-hour and task-status data.
Certification Tracker	FR-05	Implemented	Form, report and dashboard quick link.
Skill Analysis report	FR-06	Implemented	Required, possessed and missing skills with matched-skills count.
Dashboard	FR-07	Implemented	Four cards, status counts, quick links and two charts.
Interview Scheduler and date validation	FR-08	Implemented	Scheduler form and Deluge validation scripts C.1–C.3.
Company Directory	FR-09	Implemented	Company form and report.
Role-based access control	FR-10	Implemented	Administrator and Student roles with the permissions of Table 4.4.

CHAPTER 6
TESTING AND RESULTS

6.1 Testing Approach

Testing is manual and black-box: each test case exercises a requirement through the application's user interface and compares the outcome with an expected result. Fourteen test cases (TC-01 to TC-14) cover profile management, job opportunities, applications, preparation tasks, certifications, interview scheduling and date validation, skill analysis, dashboard calculations, data consistency, role-based access, form validation, and navigation and reports.

Every test case is recorded with the same attributes: test case ID, related requirement, preconditions, test steps, expected result, actual result and status. The test cases are given in Table 6.1 in Section 6.2.

6.2 Test Cases and Execution Results

Table 6.1: Test Cases and Execution Results

TC ID	Req. Ref.	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Student Profile form is open.	1. Enter name, e-mail, department, CGPA and skills. 2. Submit. 3. Edit the record and submit again.	Record is saved, listed in All Student Profiles and shows the edited values.	Record saved and listed; edited values displayed.	Pass
TC-02	FR-02	A company exists in the Company Directory.	1. Open the Job Opportunity form. 2. Enter company, role, skills, minimum CGPA and department. 3. Submit. 4. Open All Job Opportunities.	Opportunity is saved and listed with the entered values.	Opportunity saved and listed with correct field values.	Pass
TC-03	FR-03	A student and a job opportunity exist.	1. Open the Job Application form. 2. Select student and opportunity. 3. Enter status and dates. 4. Submit. 5. Open All Job Applications.	Application is saved and listed with ID, student, company, role and date.	Applications 1 to 5 listed with company, role and date; footer shows “Showing 5 of 5”.	Pass
TC-04	FR-03	An application record exists.	1. Open an application. 2. Change Application Status. 3. Save. 4. Open Application Statuses.	New status is stored and shown in the report and status view.	Updated status stored and displayed.	Pass

TC ID	Req. Ref.	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC-05	FR-04	A student exists.	1. Open the Preparation Tracker form. 2. Add a task and select a priority. 3. Submit. 4. Open the priority view.	Task appears under the selected priority.	Task listed under the chosen priority; task status and study hours feed the dashboard charts.	Pass
TC-06	FR-05	A student exists.	1. Open the Certification Tracker form. 2. Enter certification details. 3. Submit. 4. Open the report.	Certification record is saved and listed under the student.	Record saved and listed.	Pass
TC-07	FR-08	Student, company and role exist.	1. Open the Interview Scheduler form. 2. Enter a past interview date and submit. 3. Repeat with a future date.	Past date is rejected with “Interview date cannot be in the past.”; future date is accepted.	Past date rejected with the message; future date accepted.	Pass
TC-08	FR-03, FR-08	Scripts C.1 and C.3 are attached to the Job Application form.	1. Enter an Assessment or Interview Date earlier than the Application Date. 2. Submit. 3. Correct the date and submit.	Invalid submission is cancelled with the matching message; corrected record is saved.	Submission cancelled with the matching message; corrected record saved.	Pass
TC-09	FR-06	A student with skills and an opportunity with required skills exist.	1. Open All Skill Analyses. 2. Locate the student and role. 3. Compare with the source records.	Possessed and missing skills agree with the Student Profile and Job Opportunity records.	Required, possessed and missing skills and matched count agree with the source records.	Pass
TC-10	FR-07	Records exist.	1. Open the Dashboard. 2. Read the cards, status counts and charts. 3. Edit a record. 4. Refresh and compare.	Cards, counts and charts are displayed and agree with the records after refresh.	Four cards (0.00, 0.00, 83.17, 3.81), status counts and two charts displayed; values agree with the records.	Pass
TC-11	NFR-05	Students and opportunities exist.	1. Open the Job Application form. 2. Select a student and an opportunity. 3. Submit. 4. Open All Job Applications.	Student and opportunity are chosen from existing records and shown in the report.	Lookup lists show existing records; report shows the linked student names and opportunity IDs.	Pass
TC-12	FR-10, NFR-02	Administrator and Student users exist.	1. Sign in as Student; open administrator-only modules and another student's records. 2. Repeat as Administrator.	Student access is limited as in Table 4.4; Administrator has full access.	Student access limited as specified; Administrator has full access.	Pass

TC ID	Req. Ref.	Preconditions	Test Steps	Expected Result	Actual Result	Status
TC-13	FR-03, FR-08	Forms are open.	1. Submit the Job Application and Interview Scheduler forms with required fields empty. 2. Submit with invalid dates.	Submission is cancelled with a specific message for each failed rule.	Submission cancelled with the specific message for each rule.	Pass
TC-14	NFR-03	User is signed in.	1. Open each navigation section. 2. Open a report and use the search control.	Every section opens its form or report; reports show the record count.	All nine sections open; All Job Applications shows five records with search and add controls.	Pass

6.3 Functional Validation

The records in the All Job Applications report give direct evidence that applications are created and retrieved with company, role and date information (TC-03). Table 6.2 reproduces those records.

Table 6.2: Job Application Records in the Demonstration Data

Application ID	Company Name	Job Role	Application Date
1	TechNova Solutions	Software Engineer	15-Oct-2026
2	GreenLeaf Corp	Data Analyst	16-Oct-2026
3	Urban Dynamics	Project Manager	17-Oct-2026
4	SkyHigh Innovations	UX Designer	18-Oct-2026
5	Future Tech Labs	DevOps Engineer	19-Oct-2026

The dashboard presents consistent summary measures for the same data: four job-status categories each holding one record, an average CGPA of 3.81 and an average completion percentage of 83.17. Forms, reports, validation scripts and the dashboard therefore operate together on one data set, and each requirement FR-01 to FR-10 and NFR-02, NFR-03 and NFR-05 is covered by at least one passed test case.

6.4 Observations and Improvements

Review of the demonstration records and dashboard produced the four observations in Table 6.3, each with a recommended improvement for the next release.

Table 6.3: Observations and Recommended Improvements

ID	Observation	Recommended Improvement
I-01	Some Job Application records list several student names and several opportunity IDs, which shows that the lookup fields accept multiple selections.	Change both lookup fields on the Job Application form to single-select so that one record describes one student and one opportunity.
I-02	Total Salary Packages and Average of Minimum CGPA show 0.00 because the demonstration opportunities contain no values for these fields.	Enter salary and minimum-CGPA values for each job opportunity.
I-03	Company Name and Job Role are typed on the Job Application form although they are held in Job Opportunity.	Populate both fields automatically from the selected Job Opportunity.
I-04	Script C.2 rejects every past interview date, so a completed interview cannot be recorded after the event.	Apply the past-date rule only when a new interview is scheduled.

6.5 Summary of Results

Table 6.4 summarises the outcome of the fourteen test cases.

Table 6.4: Summary of Test Results

Status	Count	Test Cases
Pass	14	TC-01 to TC-14
Fail	0	—
TOTAL	**14**	

All fourteen test cases passed. The forms, reports, navigation, dashboard, validation scripts and role-based access of the application behave as specified in the requirements.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environment and Deployment

The application is hosted by Zoho Creator and is accessed through a web browser using the developer's Zoho Creator account. Continued availability depends on the account and subscription plan (NFR-04). No external integration is used. Table 7.1 summarises the deployment position.

Table 7.1: Deployment and Security Status Summary

Item	Status
Application name in Zoho Creator	Student Placement Management System (project title: Smart Placement Tracker)
Hosting	Zoho Creator cloud, accessed through a web browser
Plan	Student Edition
Workflows	Three Deluge validation scripts (Appendix C)
External integrations	None
Roles defined	Administrator and Student
Role-based permissions	Configured as in Table 4.4 and tested (TC-12)
Data	Demonstration data

7.2 User Roles and Permissions

The application defines two roles, Administrator (used for the Placement Administrator) and Student. Table 4.4 sets out the permissions of each role over the components of the application. These permissions are configured in the application and were verified by TC-12: a Student can reach only the modules and records allowed to that role, and an Administrator can reach all modules.

7.3 Data Protection and Security

The application holds student names, e-mail addresses, CGPA, skills and placement activity, which are personal and academic data (NFR-02). The following controls protect them:

- Role permissions as in Table 4.4 limit each user to the records and modules authorised for the role.
- Forms and reports are not shared through public links.
- Login credentials are not reproduced in this report.
- Demonstration records are replaced with real records before live use.

7.4 Operational Considerations

Operating the application requires keeping placement records complete and current, because dashboard and skill-analysis figures are computed from the stored data (NFR-05). The administrator maintains the company directory and job opportunities, assigns users to roles, and reviews the records against the improvements listed in Table 6.3. Changes to forms, reports and scripts are made in the Zoho Creator builder and are re-tested against the test cases of Chapter 6.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Achievement of Objectives

Table 8.1 relates each objective of Section 1.3 to what the application delivers and to the evidence for it.

Table 8.1: Mapping of Objectives to Outcomes

Objective	Reference	Outcome and Evidence
OBJ-1: Centralise student, company and job-opportunity information	Sections 4.2, 5.2; TC-01, TC-02	Student Profile, Company Directory and Job Opportunity forms and reports hold the records; students and opportunities are selectable in applications.
OBJ-2: Record and track applications	Sections 5.2, 5.3; TC-03, TC-04	Job Application form and report hold the records with status, dates and notes.
OBJ-3: Organise interviews, preparation and certifications	Sections 4.3, 5.4; TC-05 to TC-07	Interview Scheduler, Preparation Tracker (High, Medium, Low priorities) and Certification Tracker modules operate as tested.
OBJ-4: Present placement summaries	Section 5.3; TC-10	Dashboard shows four summary cards, status counts, quick links and two charts.
OBJ-5: Identify skill gaps	Sections 5.3, 5.4; TC-09	Skill Analyses report lists required, possessed and missing skills and a matched-skills count.

8.2 Analysis of Application Functionality

The application brings the placement information of a student into one interface. Records are entered through forms, retrieved through list reports and summarised on the dashboard, so a student can move from the application list to preparation tasks and skill gaps without leaving the system. The lookup fields of the Job Application form keep student and opportunity details in one place and avoid retyping them.

All ten functional requirements and the non-functional requirements NFR-02, NFR-03 and NFR-05 are covered by passed test cases (Table 6.1). The Deluge validation scripts keep date data consistent at entry, and the role configuration restricts each user

to the records the role allows. The observations of Table 6.3, particularly the multi-select lookups of I-01, identify where the data model can be tightened further.

8.3 Limitations and Discussion

The application depends on the features and limits of the Zoho Creator account and plan, in line with the platform constraints reported for low-code development generally [2]. Job opportunities are entered manually because no external data source is connected. Skill comparisons depend on accurate skill information, and dashboard values depend on complete records, as the zero-valued cards in the demonstration data show. Reminders and an activity log are future enhancements. Skill-match figures can guide preparation but do not predict recruitment outcomes, and no claim is made about changes in placement rates or user satisfaction because no such evaluation was carried out.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

Smart Placement Tracker shows that a placement-management application covering profiles, opportunities, applications, interviews, preparation, certifications and skill gaps can be built on Zoho Creator with forms, list reports, a dashboard, role-based access and a small set of Deluge validation scripts. Its main contribution is a single data-driven workspace in which application progress, preparation effort and skill gaps can be reviewed together, replacing separate spreadsheets and notes.

The project applied the course concepts of forms, reports, lookups, dashboards, scripting and role design to a realistic problem, and documented the design, the tests and the observations for the next release. The same structure can be reused for other tracking applications on the platform.

9.2 Future Enhancements

- Apply the improvements I-01 to I-04 of Table 6.3.
- Add an activity log recording who created or changed each record.
- Add automated notifications for job openings, application deadlines, assessment dates and interview reminders.
- Add resume parsing and personalised recommendations based on the requirements of a selected job role.
- Integrate with approved job portals or institutional placement data sources to reduce manual entry.
- Extend analytics to application conversion, department-wise progress, certification completion and in-demand skills.
- Improve mobile usability.

REFERENCES

[1]A. C. Bock and U. Frank, “Low-code platform,” *Business & Information Systems Engineering*, vol. 63, no. 6, pp. 733–740, Dec. 2021, doi: 10.1007/s12599-021-00726-8.

[2]Y. Luo, P. Liang, C. Wang, M. Shahin, and J. Zhan, “Characteristics and challenges of low-code development: The practitioners’ perspective,” in *Proc. 15th ACM/IEEE Int. Symp. Empirical Software Engineering and Measurement (ESEM)*, 2021, doi: 10.1145/3475716.3475782, arXiv:2107.07482.

[3]Zoho Corporation, “Deluge scripting,” *Zoho Creator Help*. [Online]. Available: https://www.zoho.com/creator/help/script/ (Accessed: Oct. 2026).

[4]Zoho Corporation, “Introduction to Deluge,” *Zoho Deluge Help*. [Online]. Available: https://www.zoho.com/deluge/help/ (Accessed: Oct. 2026).

[5]U. Frank, P. Maier, and A. Bock, *Low Code Platforms: Promises, Concepts and Prospects. A Comparative Study of Ten Systems*, ICB Research Report No. 70, University of Duisburg-Essen, 2021.

APPENDIX A – FEATURE COVERAGE SUMMARY

Table A.1 summarises the Zoho Creator features used by the project and where each is described in this report.

Table A.1: Feature Coverage Summary

Zoho Creator Feature	Status	Where in the Application	Report Reference
Application navigation and dashboard	Implemented	Dashboard and nine navigation sections	Section 5.1; Figure 5.1
Forms	Implemented	Job Application form and the other module forms	Section 5.2; Figure 5.2
Reports	Implemented	All Job Applications, Job Opportunities, Student Profiles and Skill Analyses reports	Section 5.3; Figure 5.3
Lookup fields	Implemented	Student Name and Job Opportunity on the Job Application form	Section 4.2
Preparation and certification tracking	Implemented	Preparation Tracker with priorities; Certification Tracker	Section 5.4; Table 5.2
Dashboard analytics	Implemented	Summary cards, status counts and charts	Section 5.3; Table 5.1
Deluge validation scripts	Implemented	Job Application and Interview Scheduler forms	Section 4.4; Appendix C
Users, roles and permissions	Implemented	Administrator and Student roles	Sections 4.5, 7.2; Table 4.4

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1 maps the course outcomes to the parts of the report and states how far the project demonstrates each.

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Extent Demonstrated	Self-Assessed Level	Examiner Level
CO1	Explain the principles and business value of low-code development.	Chapters 1, 2	Fully: justification, comparison and literature survey.		
CO2	Develop applications using Zoho Creator forms, reports and workflows.	Chapters 4, 5, 6	Forms, reports, lookups, dashboard and three Deluge validation scripts.		
CO3	Integrate Zoho Creator applications with other services and secure them.	Sections 4.5, 7.2, 7.3	Roles and permissions configured and tested (TC-12); access control and data protection described in Sections 4.5, 7.2 and 7.3.		
CO4	Deploy and manage applications following governance practices.	Chapter 7	Hosting environment, governance controls and operations described.		
CO5	Evaluate low-code and AI-assisted development strategies.	Sections 5.1, 8.3, Chapter 9	Application built with the Zoho Creator application tools; strengths, limitations and enhancements evaluated.		

APPENDIX C – DELUGE SOURCE CODE

This appendix contains the three Deluge validation scripts prepared for the Smart Placement Tracker. Each script runs when a form is submitted: it displays a message with the Deluge alert statement and stops the submission with cancel submit when a rule is broken. They were tested through test cases TC-07, TC-08 and TC-13 (Chapter 6). The code is laid out with consistent indentation.

Field link names used in the scripts are Student_Name, Company_Name, Job_Role, Application_Date, Assessment_Date and Interview_Date. Each script depends on these fields existing on the form it is attached to, with the names exactly as written. Scripts C.1 and C.3 both compare Interview_Date with Application_Date; if both are attached to the Job Application form the rule is applied twice, so they can be merged into one script.

C.1 Job Application Validation

Purpose: ensure that the assessment and interview dates are not earlier than the application date.

Form and trigger: Job Application; on submit (validate).

Dependencies: fields Application_Date, Assessment_Date and Interview_Date (date fields).

Expected behaviour: if a date is earlier than the application date, the message is shown and the record is not saved; otherwise the record is saved.

```
// Script ID : [DS-01]
// Location  : [Job Application -> On Validate]
// Purpose   : Validate assessment and interview dates

if (input.Application_Date != null && input.Assessment_Date != null)
{
    if (input.Assessment_Date < input.Application_Date)
    {
        alert "Assessment date cannot be earlier than the application date.";
        cancel submit;
    }
}

if (input.Application_Date != null && input.Interview_Date != null)
{
    if (input.Interview_Date < input.Application_Date)
    {
        alert "Interview date cannot be earlier than the application date.";
        cancel submit;
    }
}
```

C.2 Interview Scheduler Validation

Purpose: ensure that interview scheduling information is complete and that the interview is not set in the past.

Form and trigger: Interview Scheduler; on submit (validate).

Dependencies: fields Student_Name, Company_Name, Job_Role and Interview_Date; the system value zoho.currentdate.

Expected behaviour: a missing field or a past interview date shows the matching message and cancels the submission. See improvement I-04 in Table 6.3 for the handling of completed interviews.

```
// Script ID : [DS-02]
// Location  : [Interview Scheduler -> On Validate]
// Purpose   : Validate required scheduling details and interview date

if (input.Student_Name == null || input.Student_Name == "")
{
    alert "Please enter the student name.";
    cancel submit;
}

if (input.Company_Name == null || input.Company_Name == "")
{
    alert "Please enter the company name.";
    cancel submit;
}

if (input.Job_Role == null || input.Job_Role == "")
{
    alert "Please enter the job role.";
    cancel submit;
}

if (input.Interview_Date == null)
{
    alert "Please select the interview date.";
    cancel submit;
}
else if (input.Interview_Date < zoho.currentdate)
{
    alert "Interview date cannot be in the past.";
    cancel submit;
}
```

C.3 Record Consistency Validation

Purpose: reduce incomplete or inconsistent entries by checking required values and the order of the application and interview dates.

Form and trigger: Job Application; on submit (validate).

Dependencies: fields Student_Name, Application_Date and Interview_Date.

Expected behaviour: a missing student name or application date, or an interview date before the application date, shows the matching message and cancels the submission.

```
// Script ID : [DS-03]
// Location  : [Job Application -> On Validate]
// Purpose   : Check required values and date consistency

if (input.Student_Name == null)
{
    alert "Enter student name.";
    cancel submit;
}

if (input.Application_Date == null)
{
    alert "Select application date.";
    cancel submit;
}

if (input.Interview_Date != null &&
    input.Interview_Date < input.Application_Date)
{
    alert "Interview date cannot be before application date.";
    cancel submit;
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Sign in to Zoho Creator with an account that has been given access and open the application. It opens on the Dashboard. The menu on the left gives access to Job Applications, Job Opportunities, Preparation Trackers, Certification Trackers, Interview Scheduler, Student Profiles, Skill Analyses and Company Directory.

D.2 Instructions for Student

1. Open **Job Applications → Job Application**, select your name and the job opportunity, enter the application status and any assessment and interview dates, and submit. Dates earlier than the application date are rejected.
2. Open **Job Applications → All Job Applications** to confirm that the new record appears, and **Application Statuses** to review the stage of each application.
3. To change the stage of an application, open the record, update Application Status and save.
4. Open **Preparation Trackers**, add a task, choose its priority (High, Medium or Low) and enter planned and actual study hours.
5. Open **Certification Trackers** and add each certification you hold or are pursuing.
6. Open **Skill Analyses** to review the skills required for a job role, the skills you already have and the skills you still lack.
7. Return to the **Dashboard** to see job-status counts, completion percentage and study-hour charts.

D.3 Instructions for Placement Administrator

1. Open **Company Directory** and add or update company details.
2. Open **Job Opportunities** and add each opening with its role, required skills, minimum CGPA and eligible department.
3. Maintain **Student Profiles** and review the All Student Profiles report.

4. Review **All Job Applications** and update the application status as selection progresses.
5. Use **Interview Scheduler** to enter interview dates; a past date is rejected.
6. Review the **Skill Analyses** report and the **Dashboard** for placement indicators.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	SOWMYA R	Register No.	241501207
Project Title	Smart Placement Tracker System	Date	October 2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	SOWMYA R	Register No.	241501207
Department / Section	Artificial Intelligence and Machine Learning	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ Name: _____